(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

3rd Semester, November, 2022

Paper Code: ECC-209 **Subject: Analog Communications**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	Derive an expression for a single tone frequency modulation.	2	1, 2
1 (b)	The signal is modulated at 7.5 V and carrier voltage is of 9V, calculate modulation percentage.	2	1, 2
1 (c)	A carrier signal is frequency modulated with a sinusoidal signal of 2 KHz resulting in a maximum frequency deviation of 5 KHz. Find the bandwidth of the modulated signal.	2	1, 2
1(d)	In the demodulation process of DSB-FC using envelope detector, how the value of time constant RC determmine the distortion in received signal.	2	1, 2
1(e)	Compare the SSB-SC and DSB-SC in termis of bandwidth and power saving. Also give the practical application of VSB.	2	1, 2
			1, 2
2 (a)	Explain the Phase Locked Loop Frequency demodulator with derivation.	5	1, 2
2(b)	Differentiate between NBFM and WBFM. Where they are used? What are the merits and Limitation of FM.	5	1, 2
3 (a)	An AM transmitter radiates 9kW power when the carrier is unmodulated and 10.125kW when carrier is sinusoidally modulated. Find the modulation index. Now if another sine wave coresponding to 40 % modulation is transmitted simultaneously, calculate the total radiated power.	5	1, 2
3 (b)	Explain the suppression of carrier in generation of DSB-SC with suitable diagram using balanced Modulator.	5	1, 2
4 (a)	Explain Superhetrodyne receiver with block diagram.	5	1, 3
4 (b)	b) A carrier voltage has a peak amplitude of 10 V at a frequency of 1 MHz. A sinusoidal signal of 1 kHz varies the amplitude of RF wave between 7.5 V and 12.5 V. Find the modulation index and equation of AM wave.	5	1, 2